Dylan, Kique & Lulu

Annotated Bibliography Assignment
Research in Oceanography
Adapted from E. Halliday Walker

What is an Annotated Bibliography?

An annotated bibliography is a list of citations to books, articles, chapters, and/or other and documents. Citations are listed alphabetically according to the last name of the first author. Each citation is followed by a brief (usually about 150 words) descriptive and evaluative paragraph, the annotation.

What Purpose Do They Serve?

As a student, you will find it helpful to have an annotated bibliography as an indication of the sources you intend to use for an assignment, paper or thesis. Therefore, the purpose of writing an annotated bibliography is to record an informative/descriptive summary about the documents you read. In the future, the annotated bibliography will help remind yourself about why your cited sources are relevant to your project.

Annotations vs. Abstracts

Abstracts are the purely descriptive summaries found at the beginning of scholarly journal articles or in periodical indexes. Annotations are **descriptive** and **critical**; they expose your point of view, clarifying and highlighting the relevance of this source to your research question.

The Process

Creating an annotated bibliography calls for the application of the following intellectual skills: concise exposition, succinct analysis, and informed library research.

1. First, locate and record citations to articles from the scientific literature, books, or other documents that may contain useful information and ideas related to your research question. Briefly examine and review the actual items. For your annotated bibliography, focus on the sources that provide important background information and contextualize your research question.
2. Cite the journal article or document using the appropriate style
3. Write a concise annotation (~100-200 words) that **summarizes, assess, and reflects on the source**. If someone asked what this article is about, what would you say? How does it compare with other sources in your annotated bibliography? Are there flaws in the study design? Limitations to the author's conclusions? How has this source changed how you think about your research topic?

Assessment: I will assess the bibliographies by evaluating how thoroughly you addressed each of the 4 key components shown in the flow chart below:

-citation
-content overview
-critical analysis
-relevance

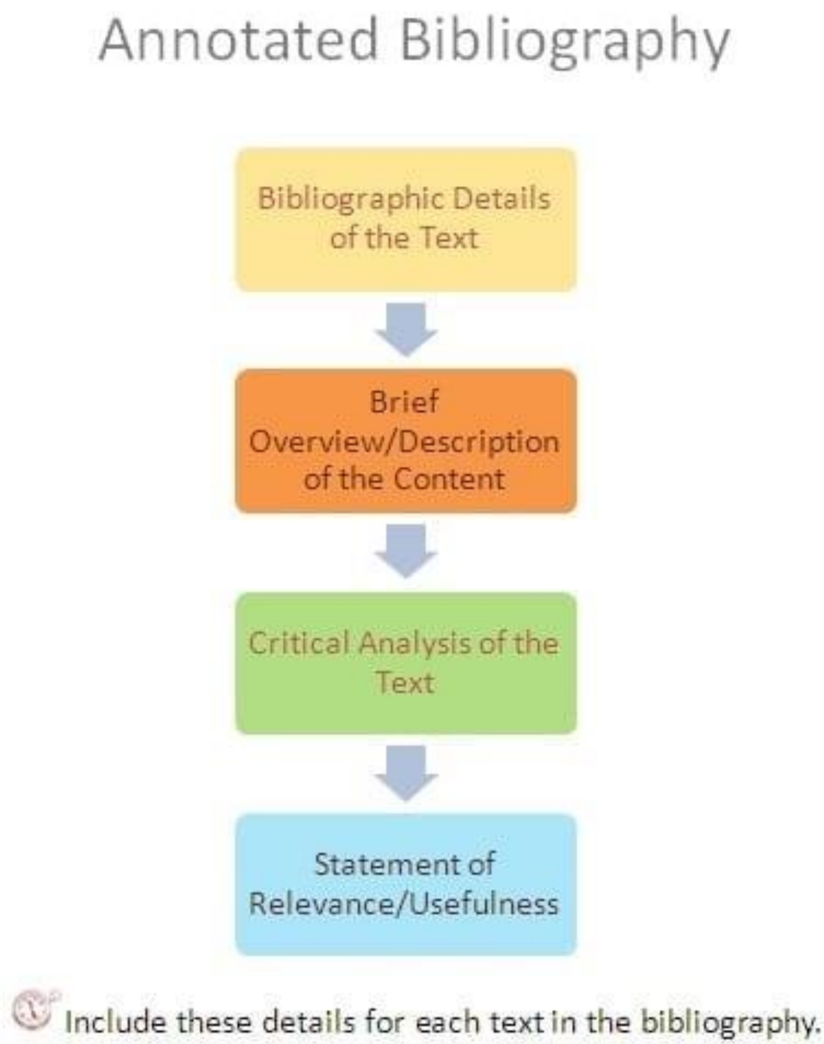

Image source: libguides.uwf.edu

Helpful Bowdoin Resources for Research and Writing:

- The library offers an excellent resource on Citation Management tools (Endnote, Zotero, EasyBib) that enable you to create a "personal library" of references, citations and documents. The library also offers and workshops and one-on-one training with these resources.

- The Center for Learning and Teaching's Writing Workshop offers one-on-one appointments for all stages of the writing process (focusing, developing, clarifying, organizing, refining).

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Rubric:

- *You will submit the first of two annotated bibliographies of <u>at least</u> 5 peerreviewed articles per person in your group from the scientific E.g. if you are working in a group of 2, summarize at least 10 articles, if you are not working in a group, summarize at least 5.*

	Name:	100%	50%	0%
portion of grade		meets criteria	partially meets criteria	not included
20%	5 references per person in group 10+ references			
10%	references primary literature			
10%	full citation included			
30%	content overview included			
30%	critical analysis/relevance included			

Arrigo, K. R., Perovich, D. K., Pickart, R. S., Brown, Z. W., Van Dijken, G. L., Lowry, K. E., Mills, M. M., Palmer, M. A., Balch, W. M., Bahr, F., Bates, N. R., Benitez-Nelson, C., Bowler, B., Brownlee, E., Ehn, J. K., Frey, K. E., Garley, R., Laney, S. R., Lubelczyk, L., … Swift, J. H. (2012). Massive Phytoplankton Blooms Under Arctic Sea Ice. *Science*, *336*(6087), 1408–1408. https://doi.org/10.1126/science.1215065

Using in-situ data, the authors document a massive phytoplankton bloom beneath fully consolidated pack ice far from the ice edge in the Chukchi Sea, where light transmission has increased in recent decades because of thinning ice cover and proliferation of melt ponds. Evidence suggests that under-ice phytoplankton blooms may be more widespread over nutrient-rich Arctic continental shelves and that satellite-based estimates of annual primary production in these waters may be underestimated by up to 10-fold. This research helps to answer the part of our research questions that looks at sea ice thickness as a key variable in primary production. Specifically, optical measurements show that light transmission through ice was enhanced by an increasing fraction of first-year ice, which is much thinner (0.5 to 1.8 m), than the historically dominant multiyear ice pack (2 to 4 m), and especially by an increasing number of surface melt ponds which transmit 4x more light than snow-free ice. While this paper is helpful in that it identifies a large portion of Arctic NPP to be under ice blooms, it will be a challenge for our team to figure out how to incorporate these blooms without in-situ cruise data.

Brown, Z. W., & Arrigo, K. R. (2012). Contrasting trends in sea ice and primary production in the Bering Sea and Arctic Ocean. *ICES Journal of Marine Science*, *69*(7), 1180–1193. https://doi.org/10.1093/icesjms/fss113

This paper contrasts sea ice and primary production trends in the Bering Sea and the Arctic Ocean. In the Arctic, diminishing sea-ice has extended the open-water growing season by 45 days in 12 years, promoting a 20% increase in NPP. This signals that light is the limiting factor in polar environments. In the Bering Sea, seasonal ice pack has remained consistent over time, partially because of winter winds, and therefore no trend in NPP was found. The implications of this research suggest that future sea-ice loss will likely stimulate additional NPP over the productive Bering Sea shelves, potentially reducing nutrient flux to the downstream western Arctic Ocean. This paper is of particular relevance due to their use of satellite derived chlorophyll a, sea surface temperature, and sea-ice cover observations from 1998 through 2007. To characterize spatial differences, the authors subdivided the Arctic Ocean into eight geographic sectors by longitude (see methods) which may be a relevant tool for our own analysis.

Castagno, A. P., Wagner, T. J. W., Cape, M. R., Lester, C. W., Bailey, E., Alves-de-Souza, C., York, R. A., & Fleming, A. H. (2023). Increased sea ice melt as a driver of enhanced Arctic phytoplankton blooming. *Global Change Biology*, 29, 5087–5098. **https://doi.org/10.1111/gcb.16815**

This study analyzes how sea ice melt influences phytoplankton blooms. They find a general increase in blooms as sea ice melts as one might expect. There is a clear light limitation as in some light limited areas the loss of ice stimulates large blooms. However, light availability and productivity were not perfectly correlated. This implies that light is not the main driver at play. Notably, melting ice injects cold fresh water into the surface water and causes mixing. This mixing can then cause nutrients to be brought to the productive surface ocean in the Arctic. Therefore, the areas with the biggest blooms as a result of sea ice melt were in the interior Arctic where these density differences can be at play.

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Castellani, G., Veyssière, G., Karcher, M. *et al.* Shine a light: Under-ice light and its ecological implications in a changing Arctic Ocean. *Ambio* **51**, 307–317 (2022). https://doi.org/10.1007/s13280-021-01662-3

This paper models light attenuation through sea ice. Using their model they are able to understand how light penetrates through light. At a basic level ice degrades light about 10 times more quickly than ocean water, but it is more complex. Depending on the snow cover and melting patterns light penetration can change. However, through their model they are able to estimate PAR under sea ice and thus some primary productivity. They also note how some phytoplankton species have adapted to thrive at low light levels under the ice and still tolerate significantly higher levels during the summer months when no ice is present.

Douglas, C. C., Briggs, N., Brown, P., MacGilchrist, G., & Naveira Garabato, A. (2024). Exploring the relationship between sea ice and phytoplankton growth in the Weddell Gyre using satellite and Argo float data. *Ocean Science*, *20*(2), 475–497. https://doi.org/10.5194/os-20-475-2024

This article investigates the relationship between sea ice and net primary productivity (NPP) in the Antarctic Weddell Gyre from 2002-2020. They combine satellite NPP products (CAFE model) with in-situ data (11 Argo floats) to track chl-a and particulate organic carbon (POC). They conclude that summer maximum ice free area is the strongest predictor of annual NPP, and while productivity is higher on the continental shelf, the massive open-ocean region accounts for 93-96% of the basins total annual carbon uptake. Unlike Xi et al. (2021), this article conducts a regional analysis on NPP, not specific to PFTs. While researching opposite poles, this study shares a similar focus to Frey et al. (2023), regarding the impact of ice-free days on productivity. While Arrigo et al., (2012) emphasizes under-ice blooms in the Arctic, this article finds that the majority of phytoplankton biomass in the Weddell Gyre is found in ice-free open ocean, with only 10% of annual chl-a found under ice. The Argo floats also lacked Photosynthetically Active Radiation (PAR) sensors which prevented the authors from directly calculating water-column integrated NPP from in-situ data. A significant limitation is that the annual NPP estimates are not independent of the ice-free area, because the satellite algorithms default NPP to zero under ice, which is something we should integrate into our analysis. This source provides a methodological template for integrating autonomous in-situ measurements with satellite time-series data to measure NPP in polar regions. However, it highlights a clear knowledge gap for the Arctic, which is the need to combine this integrated approach with PFT-type products (like GLOPHYTs) and sea-ice thickness data.

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Horvat C, Bisson K, Seabrook S, Cristi A and Matthes LC (2022) Evidence of phytoplankton blooms under Antarctic sea ice. Front. Mar. Sci. 9:942799. doi: 10.3389/fmars.2022.942799

This paper utilizes a model to simulate under ice blooms (UIBs). Just like we plan to, the leveraged data from in situ ARGO floats to design their model. This also allowed them to get around remote sensing data not being able to measure underneath ice. With these observations and their model Horvat et al. were able to forecast the likelihood of an UIB throughout the year in Antarctica. These calculations include a mix of calculating under ice PAR and in situ measurements of nutrient availability, temperature, and stratification. Altogether this study gives us a framework for analyzing under ice phytoplankton growth that should be useful in our project.

Kahru, M., Brotas, V., Manzano-Sarabia, M., & Mitchell, B. G. (2011). Are phytoplankton blooms occurring earlier in the Arctic? *Global Change Biology*, *17*(4), 1733–1739. https://doi.org/10.1111/j.1365-2486.2010.02312.x

This team created a time series of satellite-derived surface chlorophyll-a concentrations from 1997–2009 to analyze trends in the timing of the annual phytoplankton bloom maximum. They found that the annual summary phytoplankton bloom maximum has advanced by up to 50 days which coincides with areas where ice concentration has decreased in early summer, thus making the earlier blooms possible. This is key finding for the relevance of our project; it’s important to study the ecosystem implications of changes to sea-ice extent and timing associated with climate change. This study also uses a unique methodology for studying primary productivity where instead of quantifying carbon usage, POC, or community composition, they use the chlorophyll surface maximum from remote sensing data to track trends in the timing of the blooms across a multiyear study period. We may consider this methodology to track how changes to phytoplankton bloom timing relate to key sea-ice parameters.

Lenss, Megan, Moreau, Sebastien, Hattermann, Tore, Wiktor, Jozef, Różańska, Magdalena, Claeys, Philippe, Brion, Natacha, Chierici, Melissa, Fransson, Agneta, Campbell, Karley; Phytoplankton bloom distribution and succession driven by sea-ice melt in the Kong Håkon VII Hav. *Elementa: Science of the Anthropocene* 12 January 2024; 12 (1): 00122. doi: https://doi.org/10.1525/elementa.2023.00122

This study looks at the phytoplankton concentration at the ice edge as that interface moves in the Antarctic. They find higher chlorophyll concentrations at these ice edge areas as the recent melt leads to more light being available and an injection of cold fresh water causing mixing and thus an increase in nutrients. Additionally this study looks at community composition. The community composition of these ice edge areas are far more representative of an open ocean community than others underneath the ice. That being said there are still more flagellates and less centric and pennate diatoms than we would expect in an open ocean environment.

Lester, C. W., Wagner, T. J. W., & McNamara, D. E. (2025). A model of near-sea ice phytoplankton blooms. *Limnology and Oceanography Letters*, *10*(2), 245–253. https://doi.org/10.1002/lol2.10449

This study develops a biophysical model to explain how sea ice meltwater drives the magnitude and spatial distribution of phytoplankton blooms near the ice edge. The study demonstrates that meltwater creates a shallow, stratified surface mixed layer that facilitates rapid exponential growth by keeping phytoplankton cells within sunlit waters. By comparing the model to a decade of satellite observations from the Fram Strait, the authors show these blooms peak roughly 100km from the ice edge before decaying as vertical mixing deepens the mixed layer. This rouce highlights the specific role of meltwater flux and stratification in creating favorable growth conditions, whereas Douglas et al., (2024) and Frey et al., (2023) emphasized total number of "ice free days" as a driver of productivity. The study uses an "idealized" model design that purposefully simplifies the marine system, which leads to the omission of nutrient dynamics and mesozooplankton predation. Also, by assuming nutrients are not limited in early Spring, and that grazing is negligible the model may lack resolution to accurately predict the termination or species succession of the bloom. A significant limitation is that the model results are specific to the Fram Strait, which has unique current patterns that might not represent other Arctic regions like the Pacific Arctic. This article also lacks integration of in-situ data and is constrained by sea surface measurements. This paper emphasizes that the Spring blooms in the Arctic are not just about the influx of light or nutrients, but also the physical stability of the water column, and mixed layer depth. It highlights the need to link sea ice thickness and melt rates, to the taxonomic composition of the bloom. It also reinforces the idea that future sea ice retreat won't simply increase production linearly but instead might fundamentally suppress traditional near-ice blooming patterns as the stratified meltwater layer disappears.

Lewis, K. M., Van Dijken, G. L., & Arrigo, K. R. (2020). Changes in phytoplankton concentration now drive increased Arctic Ocean primary production. *Science*, *369*(6500), 198–202. https://doi.org/10.1126/science.aay8380

This article found that the Arctic Ocean net primary productivity (NPP) increased by 57% from 1998 to 2018, a trend initially driven by sea ice loss, but more recently by a massive increase in phytoplankton biomass. They use a specialized ocean color algorithm parameterized for the Arctics unique optical properties to show that the inflow shelves contributed to 70% of this total increase. They conclude that an influx of new nutrients is likely sustaining this growth as the

Arctic transitions from being light-limited to being potentially nutrient-driven. Like Frey et al. (2023) and Douglas et al. (2024), this study highlights the critical role of ice-free area in productivity, yet it uniquely identifies a decadal shift where phytoplankton concentration has overtaken open-water area as the primary driver of NPP variance. The study design relies satellite-derived surface chlorophyll, which can underestimate NPP by missing massive under-ice and deep-water blooms. They also highlight that by using bulk chl-a rather than PFT products, they cannot analyze community composition. A major limitation is that they infer added nutrients through statistical correlation rather than direct measurement of nutrient concentrations or fluxes. Their findings are also highly regional, so the trend they found is not pan-Arctic. Additionally, the "slowing" of sea ice decline observed in the second decade may simply be internal climate variability masking long-term trends, which could complicate the conclusion that drivers have fundamentally shifted. This fundamental shift, from a “white ocean” (ice-dominated) to a “green ocean” (nutrient + biomass-dominated) will be interesting to analyze, in our study, with the added data layers of nutrient concentration measurements from in-situ data and community composition analysis from remote sensing data.

Oziel, L., Baudena, A., Ardyna, M., Massicotte, P., Randelhoff, A., Sallée, J.-B., Ingvaldsen, R. B., Devred, E., & Babin, M. (2020). Faster Atlantic currents drive poleward expansion of temperate phytoplankton in the Arctic Ocean. *Nature Communications*, *11*(1), 1705. https://doi.org/10.1038/s41467-020-15485-5

This article investigates the “Atlantification” of the Arctic Ocean, specifically how strengthening of North Atlantic currents facilitates the poleward expansion of temperate phytoplankton species. The study focuses on the coccolithosphere, *Emiliania huxleyi* (*Ehux)*, as a tracer for temperate ecosystems, using satellite-derived surface currents and Particulate Inorganic Carbon (PIC) as a biomass proxy. They find that bio-advection, rather than just increasing water temperatures, is the primary mechanism driving this species into the Barents Sea. This study conducts a regional analysis of a single indicator species, and the physical forcing behind its movement, which differs from studies looking at bulk chl-a or PFTs. Frey et al. (2023) and Douglas et al. (2024) examine how sea ice retreat influences local NPP, whereas this study emphasizes the role of horizontal transport (advection) in changing community composition, which is driven by a migration of a southern species, northward. One limitation is that the reliance on PIC as a proxy for *Ehux* is proxy-based rather than a direct taxonomic identification. This study is also regionally specific, the same physical processes driving this phenomenon might not apply to other Arctic regions. This source adds an important dimension to our research, by demonstrating that Arctic community composition is not just a product of local environmental changes. It highlights that current-driven and bio-advection can import non-indigenous PFTs, fundamentally altering the regional food web.

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Pabi, S., Van Dijken, G. L., & Arrigo, K. R. (2008). Primary production in the Arctic Ocean, 1998–2006. *Journal of Geophysical Research: Oceans*, *113*(C8), 2007JC004578. https://doi.org/10.1029/2007JC004578

This paper develops an algorithm for primary production that uses remotely sensed chlorophyll a, sea surface temperature, and sea ice extent data. They apply their algorithm to quantify interannual changes in phytoplankton production in the Arctic Ocean between 1998 and 2006. Generally, the results show annual production was roughly equally distributed between pelagic waters (less productive but greater area) and waters located over the continental shelf (more productive but smaller area). The authors also show that interannual differences in primary production are most tightly linked to changes in sea ice extent, with changes in sea surface temperature (related to the Arctic Oscillation) playing a minor role. This paper's methodology will be key in how our group decides to quantify primary production in the Arctic. While many papers use chlorophyll a proxies inferred from ocean color, this paper develops an algorithm that computes daily primary productivity in units of (mg C /m^2 / d). The inputs of this algorithm are reviewed in depth in the methods section but can be summarized as: *surface Chl a concentrations* determined from SeaWiFS ocean color data (and excluding river influenced pixels); *sea surface temperature* based on the Reynolds Optimally Interpolated SST (OISST) Version 2 product obtained from NOAA; and *open water area* estimated from Special Sensor Microwave Imager. Becoming familiar with both the sources of data and how to use them within the context of our study is an important first step for when we dive into the analysis component of our research.

Serreze, M. C., Holland, M. M., & Stroeve, J. (2007). Perspectives on the Arctic's Shrinking Sea-Ice Cover. *Science*, *315*(5818), 1533–1536. https://doi.org/10.1126/science.1139426

They key finding of this paper is that arctic sea-ice is shrinking; linear trends in arctic sea-ice extent over the period 1979 to 2006 are negative in every month. While this isn't a huge development in our team's understanding the state of the Arctic Ocean, it's fleshes out one half of our research goal which is to find out how sea-ice extent and thickness are changing over time. More importantly, it discusses mechanism for this observed ice loss which informs our fundamental understanding of the Arctic system. The authors propose that a combination of strong natural variability in the coupled ice-ocean-atmosphere system and a growing radiative forcing associated with rising concentrations of atmospheric greenhouse gases explains most of the ice-loss. The paper also overviews changes to ice circulation associated with the behavior of the North Atlantic Oscillation and Northern Annular Mode which are good climate variables to be aware of for our project. Lastly, this paper looks at trends in ice extent in each month which gives a more complete picture of Arctic sea-ice than most of our other background literature which, by virtue of studying primary productivity, only reports ice extent in the summer months.

Smith, W. O.Jr., and J. C. Comiso (2008), Influence of sea ice on primary production in the Southern Ocean: A satellite perspective, *J. Geophys. Res.*, 113, C05S93, doi:10.1029/2007JC004251.

This paper discusses trends in primary productivity and sea ice cover in the Southern Ocean. Specifically Smith et al. find have found steady increases in production recently, especially in the summer months. They acknowledge that this would appear to be a product of decreased ice cover. However, when comparing their different sites, they have good evidence to support that this likely has more to do with decreased cloud cover and increased iron availability which also could be caused by a changing climate. This paper also does a great job of acknowledging the limitations of their modeling approach. In particular, they note that there is small but significant production underneath sea ice that they are unable to account for. This is a gap we will definitely want to explore in our project.

Xi, H., Losa, S. N., Mangin, A., Garnesson, P., Bretagnon, M., Demaria, J., Soppa, M. A., Hembise Fanton d'Andon, O., & Bracher, A. (2021). Global Chlorophyll *a* Concentrations of Phytoplankton Functional Types With Detailed Uncertainty Assessment Using Multisensor Ocean Color and Sea Surface Temperature Satellite Products. *Journal of Geophysical Research: Oceans*, *126*(5), e2020JC017127. https://doi.org/10.1029/2020JC017127

This article is about the global retrieval of phytoplankton functional types (PFTs), including diatoms, haptophytes, prokaryotes, and dinoflagellates. The method employs empirical orthogonal functions (EOFs) on spectral remote sensing reflectance to reduce the high dimensionality of satellite spectral data and assess the dominant signals related to phytoplankton pigments. This study integrates data from the GlobColour merged ocean color products, combining observations from sensors like SeaWiFs, MODIS-Aqua and MERIS. Unlike regional studies such as Douglas et al., (2024) or Frey et al., (2023), which focus on the relationship between sea ice and total net primary productivity in the Antarctic Weddell Gyre and Pacific Arctic, this study provides a foundational global methodology and dataset for identifying specific community compositions, including those in the Polar regions. While Kahru et al. (2011) and Arrigo et al. (2012) focus on the timing and presence of blooms, this article provides a tool to distinguish who is blooming, rather than just where and when. The per-pixel predictive uncertainty and R^2 varies with PFT, this will be important to keep in mind if we chose to use this dataset in our analysis. Another limitation is that the study relies on Diagnostic Pigment Analysis (DPA), which carries inherent uncertainties because different taxonomic groups often share marker pigments, potentially leading to misassignment of PFTs. They highlight a knowledge gap concerning the specific application and analysis of PFT-retrieval methods in the Arctic, emphasizing that Arctic research often relies too heavily on bulk chl-a and NPP data. The publicly available GLOPHYTS dataset allows our research to gain a more robust understanding of how sea ice thickness and extent controls not just the magnitude of NPP, but also the taxonomic composition of the Arctic ecosystem.